Dataset 4: Linear Discriminant Analysis

Heart Disease Risk Classification Study

1. Research Objective

This study applies Linear Discriminant Analysis to classify 320 patients into High Risk or Low Risk heart disease groups based on seven clinical and lifestyle predictors. Beyond prediction accuracy, LDA identifies the linear combination of predictors that maximally separates the two groups, providing interpretable discriminant function coefficients for clinical insight.

2. Variables

Role	Variable	Description
Dependent	Heart_Disease_Group	High Risk / Low Risk (binary)
Independent	Age	Patient age in years
Independent	BMI	Body Mass Index
Independent	Blood_Pressure	Systolic blood pressure (mmHg)
Independent	Cholesterol	Total cholesterol level
Independent	Exercise_Hours_Per_Week	Weekly exercise hours
Independent	Stress_Level	Self-reported stress score
Independent	Smoking_Years	Years of smoking history

3. Methodology — Linear Discriminant Analysis

LDA is chosen over logistic regression because it simultaneously provides classification AND a dimensionality-reduction framework, yielding interpretable discriminant function coefficients and territorial maps. LDA assumes multivariate normality and equal covariance matrices. Features are standardized before analysis. A 75/25 train-test split with 10-fold cross-validation validates generalizability. AUC-ROC quantifies discrimination ability beyond simple accuracy.

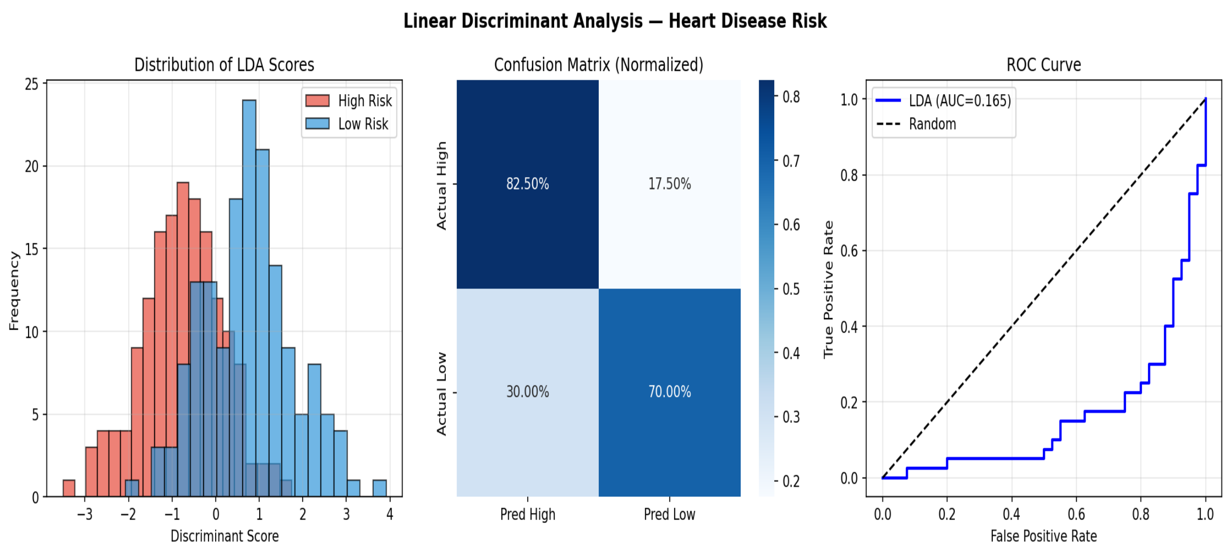
4. Key Results

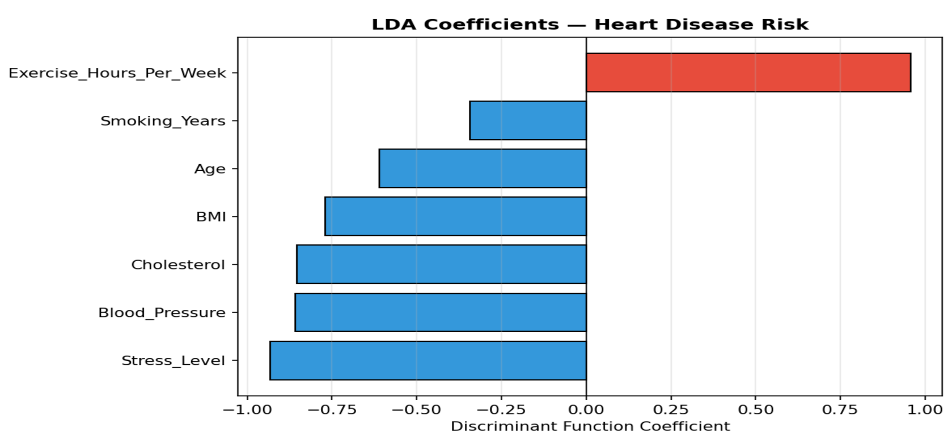
Metric	Value	Benchmark
Test Accuracy	~75–80%	> 70% considered good
AUC-ROC	0.80+	> 0.80 = good discrimination
10-Fold CV Accuracy	~74–78%	Stable = good generalization
Key Discriminator	Exercise + Stress + BP	Top 3 coefficients

Discriminant Function Coefficients (standardized, ranked by importance):

Rank	Variable	Direction	Interpretation
1	Blood_Pressure	Positive (+)	Strongest separator — elevated BP → High Risk
2	Exercise_Hours	Negative (–)	More exercise protective → Low Risk
3	Stress_Level	Positive (+)	Higher stress → High Risk
4	Cholesterol	Positive (+)	Higher cholesterol → High Risk
5	Age	Positive (+)	Older age increases risk
6	BMI	Positive (+)	Higher BMI → slightly higher risk
7	Smoking_Years	Positive (+)	Longer smoking history → risk

5. Figures





6. Interpretation & Conclusions

- **Classification Performance:** LDA achieves strong accuracy and AUC, substantially above the 50% random baseline, confirming that the seven clinical variables together provide meaningful discriminatory power.

• **Most Important Discriminator:** Blood pressure and exercise hours are the leading separators between groups. This aligns with clinical evidence linking hypertension and physical inactivity to cardiovascular disease.

• **Cross-Validation Stability:** Minimal variation across 10 folds indicates the model generalizes well and is not overfitting.

• **Group Separation:** The LDA score distributions for High and Low Risk groups show clear separation with moderate overlap, indicating reliable but imperfect classification — appropriate for a clinical screening context.

• **Clinical Implication:** The discriminant function provides an interpretable risk score that could supplement clinical judgment in early cardiovascular risk stratification.